ETALE LOCALLY CONSTRUCTIBLE SUBSETS ARE EXACTLY THE LOCALLY CONSTRUCTIBLE SUBSETS OF THE TOPOLOGICAL SPACE

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Editorial note. The stratification input is not carded (its statement carries a reference block with a citation the chapter cannot resolve) and is declared external.

Allowed standard prerequisites (editorial).

- Retrocompact equals quasi-compact opens. Topology, lemma topology-quasi-separated-scheme (cross-chapter, named). Used in the forward direction.
- Inverse images of constructibles. Topology, lemma inverse-images-constructibles (cross-chapter, named). Forward direction finish.
- Finite quasi-separated stratification. lemma-filter-quasi-compact-quasi-separated (same chapter, NOT carded, declared external). The stratification used in the converse direction.
- Constructible sets in qcqs schemes. Properties, lemma constructible-quasi-compact-quasi-separated (cross-chapter, named). E_p locally constructible implies constructible in V_p.
- Intersections and collation of constructibles. Topology, lemmas intersect-constructible-with-closed and collate-constructible-from-constructible (cross-chapter, named). Assemble E from the strata pieces.

Source reference. Original lemma 07ST: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1889–1889. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 054E: focused-seeds/geometry/S9-decent-locally-constructible/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ECU: focused-seeds/geometry/S9-decent-locally-constructible/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 005G: focused-seeds/geometry/S9-decent-locally-constructible/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted

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Source reference. Original reference 09YK: focused-seeds/geometry/S9-decent-locally-constructible/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09YE: focused-seeds/geometry/S9-decent-locally-constructible/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 005I: focused-seeds/geometry/S9-decent-locally-constructible/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0069: focused-seeds/geometry/S9-decent-locally-constructible/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-separated algebraic space over S. Let $E \subset |X|$ be a subset. Then E is étale locally constructible (Properties of Spaces, Definition 0ECU) if and only if E is a locally constructible subset of the topological space $|X|$ (Topology, Definition 005G).*

Proof. Assume $E \subset |X|$ is a locally constructible subset of the topological space $|X|$. Let $f : U \to X$ be an étale morphism where U is a scheme. We have to show that $f^{-1}(E)$ is locally constructible in U. The question is local on U and X, hence we may assume that X is quasi-compact, $E \subset |X|$ is constructible, and U is affine. In this case $U \to X$ is quasi-compact, hence $f : |U| \to |X|$ is quasi-compact. Observe that retrocompact opens of $|X|$, resp. U are the same thing as quasi-compact opens of $|X|$, resp. U, see Topology, Lemma 0069. Thus $f^{-1}(E)$ is constructible by Topology, Lemma 005I.

Conversely, assume E is étale locally constructible. We want to show that E is locally constructible in the topological space $|X|$. The question is local on X, hence we may assume that X is quasi-compact as well as quasi-separated. We will show that in this case E is constructible in $|X|$. Choose open subspaces

$$\emptyset = U_{n+1} \subset U_n \subset U_{n-1} \subset \ldots \subset U_1 = X$$

and surjective étale morphisms $f_p : V_p \to U_p$ inducing isomorphisms $f_p^{-1}(T_p) \to T_p = U_p \setminus U_{p+1}$ where V_p is a quasi-compact separated scheme as in Lemma 07ST. By definition the inverse image $E_p \subset V_p$ of E is locally constructible in V_p. Then E_p is constructible in V_p by Properties, Lemma 054E. Thus $E_p \cap |f_p^{-1}(T_p)| = E \cap |T_p|$ is constructible in $|T_p|$ by Topology, Lemma 09YE (observe that $V_p \setminus f_p^{-1}(T_p)$ is

quasi-compact as it is the inverse image of the quasi-compact space U_{p+1} by the quasi-compact morphism f_p). Thus

$$E = (|T_n| \cap E) \cup (|T_{n-1}| \cap E) \cup \ldots \cup (|T_1| \cap E)$$

is constructible by Topology, Lemma 09YK. Here we use that $|T_p|$ is constructible in $|X|$ which is clear from what was said above. □